

Luke Brzozowski

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Result-driven with 3+ years work experience in Engineering, seeking positions in Embedded Systems or Software Engineering.

EDUCATION

The University of Michigan - Ann Arbor | *Bachelor of Science in Engineering in Computer Engineering* Dec 2025
Courses: **Computer Architecture**, Embedded System Design, Embedded Control, Signals and Systems, Digital Logic Design, Computer Security, Computer Vision, Computational Linear Algebra, Data Structures and Algorithms

EXPERIENCE

AgenticFruit/isba.ai, *Technical Co-Founder* | Ann Arbor, MI May 2025 – Present

- Built multi-agent AI coding system (**TypeScript/LangGraph/Next.js**) with 90% code accuracy and 60% dev time reduction
- Implemented RAG system using Claude, Pinecone, and prompt engineering techniques, achieving 92% user query satisfaction
- Developed AI hotel administration platform with automated full-stack code generation (QlApps/Stripe/ElevenLabs integration)

Robert Bosch, *Systems Engineering Intern* | Plymouth, MI May 2024 – Jan 2025

- Led end-to-end development of embedded software for voice-activated ADAS using **C++** and **Raspberry Pi**, optimizing Google Dialogflow for 90% intent classification accuracy and 200ms response time with reliable vehicle integration via CAN bus. Delivered market-ready system showcased at CES 2025 to 10,000+ attendees, receiving positive feedback from automotive executives
- Introduced and developed automation scripting solutions using **Python** (Selenium, pandas) and Excel scripting to streamline data collection, processing, and bulk database updates, resulting in 90% reduction in manual labor for associated tasks

UTAC Inc., *Software Development Intern* | Northville Township, MI May 2023 – Aug 2023

- Utilized LabVIEW to enhance software functionality by integrating email, data acquisition systems, and data export features
- Developed live power monitoring and logging system for test stands with cost reporting module for usage analytics

UoM Formula SAE Electric, *Autonomous Team Member* | Ann Arbor, MI Aug 2023 – Jan 2024

- Developed trajectory control system for autonomous vehicle, increasing tracking accuracy by 25% over previous versions

UoM Formula SAE Combustion, *Powertrain Team Member* | Dearborn, MI Aug 2022 – Apr 2023

- Designed and simulated parts in SOLIDWORKS, defined requirements, manufactured, and validated through testing
- Rebuilt engine and conducted dynamometer analysis for performance tuning

PROJECTS

Out-of-Order RISC-V Microprocessor | *SystemVerilog, RISC-V, Python, System Verification*

- Designed 32-bit out-of-order RISC-V processor achieving 12.87ns critical path and 15% IPC improvement over normal 5-stage pipeline. Developed verification framework using **SystemVerilog** testbenches and Python automation
- Constructed modular and configurable GUI visual debugger based on Python (Tkinter) and VCD file parsing

Motion-Sensed Alarm System | *PIR Sensors, ESP32, IoT, Tinkercad, Rapid Prototyping*

- Developed IoT security system with PIR detection, wireless alerts, and live video streaming, achieving 98% detection accuracy
- Designed waterproof housing with CAD and 3D printing, using iterative prototyping to deliver durable, weather-resistant system

Accelerometer Data Logger | *C++, Arduino IDE, PCB Design, Microcontroller Programming*

- Designed and built **ESP32**-based PCB for 3-axis acceleration logging and real-time upload to Google Sheets for data acquisition

Piazza Post Categorizer | *Machine Learning (ML), C++, Binary Search Tree*

- Developed ML-based classifier for online forum posts (Piazza), achieving 90% accuracy on labeled datasets

NASA Mars Rover (Prototype) | *C++, Arduino Uno, Arduino IDE, 3D Printing*

- Led team of 4 in prototyping model of NASA Mars Rover within design constraints using rapid prototyping, engineering design standards, and requirements management. Implemented collision avoidance and functional robotic arm systems
- Designed, 3D printed, and tested multi-terrain wheels optimized for stability and traction across various surface conditions

LEADERSHIP

National Gaming Competition

- Founded & led 120-member team in securing 2nd place in three-month global gaming competition among hundreds of teams

Student Leader

- Led community service event as Student Leader, Early College Alliance at Eastern Michigan University

TECHNICAL SKILLS

Languages	C, C++, Python, SystemVerilog, ARMv7-M Assembly, Bash, SQL, TypeScript, JavaScript, MATLAB, OOP
Embedded	Raspberry Pi, ESP32, STM32Nucleo, FreeRTOS, Linux (Arch, Ubuntu), RTL Design, CAN bus
Tools	Git, Docker, Quartus Prime, Arduino IDE, Verdi, logic analyzer, SolidWorks, Simulink, oscilloscope, soldering
Other	Systems integration, debugging, validation, real-time analysis, leadership, algorithm optimization, documentation